FOR ENGINEERS — THE LEDGER AN EPC CHECKS FIRST

THE ABSENCE AUDIT — THE METHOD, AT NATIONAL SCALE

# Tunisia, audited — the engineer's ledger

the energy balance, water chemistry, acid-stream assays, lift floors, and every gate — everything a technical team verifies before FID.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

## 1. The energy ledger, leg by leg

Recovery ledger recomputed: 33.9 GWh/yr recovered + 20 GWh/yr dispatch value. The printed 82 GWh did not sum from its own legs (7.2 + 30 + 20 = 57.2; 24.8 unexplained) — corrected, with the arithmetic, on the page.

| LEG | FORMULA | VALUE |
|---|---|---|
| Pump-as-turbines | 9.81 × (70% × 150,000/86,400) m³/s × 50 m × 75% × 8,760 h | 3.9 GWh/yr |
| Heat recovery | 0.6 GJ/t × 2 Mt PG → 333 GWh thermal × 9% | 30 GWh/yr |
| Flexibility | desal + storage follow the solar curve | 20 GWh/yr (dispatch, not joules) |
| ERD | inside the 4 kWh/m³ SWRO benchmark (~35-40% of RO energy) | 0 added — no double count |

## 2. Water — the numbers an EPC checks first

| PARAMETER | VALUE | ANCHOR |
|---|---|---|
| SWRO specific energy | 4 kWh/m³ (benchmark incl. ERD) | Rw3 (mega-plant class) |
| Lift physics floor | 1.09 kWh/m³ per 400 m at 100% eff; 1.45 at 75% | Rw4 — the physics that decides where water may travel |
| Delivered inland cost | $1.10-1.50/m³ vs irrigation tariffs $0.04-0.07 | Rw6/Rw7 — inland water runs on brackish, not long-haul |
| SWRO CAPEX | $800-1,200 per m³/day (mega-plant) | Rw2 |
| Gate | blended ≥$0.68/m³, ≥60% pre-sold, brine permit | computed (DSCR) |

## 3. Materials — the acid stream

| PARAMETER | VALUE | ANCHOR |
|---|---|---|
| REE content, Gafsa rock | 322 mg/kg avg, peaks >1,700 | Rm1 |

| PARAMETER | VALUE | ANCHOR |
|---|---|---|
| PG partition of REE | 60-85% | Rm2 |
| Acid-stream U | 85-90% of rock-U partitions to the wet-process acid | Ru1 (Freeport precedent) |
| U recovery (proven) | >95% DEHPA/TOPO, 14M lbs over 1978-99 | Ru1 |
| Radiology | Ra-226 214-970 Bq/kg in PG; exemption threshold 1,000 Bq/kg | Rm11/Rm12 |
| Phase 0 assays | 12 stream points: U, REE, Ra-226, Po-210, full metals | itemized in the audit |

## 4. The dependencies, by status

| DEPENDENCY | STATUS |
|---|---|
| Desal brine permit | **in design** — Gabes outfall / diffuser; permit required before SWRO FID |
| PG chemistry (Ra-226 partitioning) | **Phase 0 assay** — flowsheet keeps radium in gypsum (Freeport precedent); re-verify on Gafsa stream |
| Uranium handling | **gate** — CNSTN licensing, NORM transport, export controls |
| REE separation capex | **later phase** — +$244M solvent-extraction plant beyond the ~$100M circuit |
| Solar bankability | **gate** — DSCR ~0.9-1.0 at the record-low tariff; bankable region or grant support required |
| Grid connection | **later phase** — 3 GW connection + 400 kV upgrades ~$450M |
| Compute accelerators | **gate** — export-control licenses; colocation until an anchor tenant is licensed |
| Truffle yield | **gate** — 376 kg/ha is the 20-yr Murcia average; gate = 3-season record >=200 kg/ha |
| Export prices | **exposure** — U spot $80-95/lb; truffle EUR20-40/kg — no contracted offtake yet |
| Financing | **uncommitted** — DFI channel modeled for solar/desal; nothing committed |

**What this document is:** a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

**What it is not:** a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

### Works cited


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2. [Rm2] Global phosphogypsum: generation, stockpiles and 60-85% REE partition to PG — https://www.mdpi.com/2071-1050/17/19/8828...
3. [Rm3] Gabès phosphate plants REE characterization: enrichments to 1,075 mg/kg; Ce 80-309 mg/kg; negligible Dy/Tb — https://www.researchgate.net/publication/352414505_Rare_earth_elements_characterization_associated_to_the_phos...
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8. [Rm17] GCT throughput (~6.5 Mt phosphate rock/yr; >10 Mt/yr PG at peak; 'hundreds of millions of tons' discharged at Gabès) — http://www.gct.com.tn/fileadmin/user_upload/Downloads/Rapport_Annuels/GCT-ANNUAL-REPORT-2012-PART1.pdf...
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10. [Rw2] SWRO CAPEX $800-1,200 per m³/day for mega-plants (>100k m³/d) — https://supra-international.com/insights/seawater-reverse-osmosis-swro-desalination-technology-comprehensive-a...
11. [Rw3] LCOW at plant gate: $0.40-0.60/m³ (Jubail 3A $0.41; Sorek II bid $0.32-0.45); energy share 40-60% of OPEX — https://pmc.ncbi.nlm.nih.gov/articles/PMC8953854/...
12. [Rw4] Pumping physics: 400 m lift = 1.09 kWh/m³ at 100% eff; 1.45 at 75%; friction 0.5-1.5 kWh/m³ over 300 km; realistic 2.0-2.5 kWh/m³ — https://www.engineeringtoolbox.com/water-pumping-costs-d_1527.html...
13. [Rw5] Jubail-Buraydah IWTP: 587 km, 650,000 m³/d, $2.26B ($3.85M/km); twin-pipe scale $5-7M/km — https://www.meed.com/saudi-arabia-awards-22bn-water-transmission-contract...
14. [Rw6] Delivered inland desalinated cost $1.10-1.50/m³ (plant gate + transport + pumping energy) — https://investriyadh.ai/intelligence/water-scarcity-investment/...
15. [Rw7] Tunisian water tariffs: SONEDE 0.200-2.310 TND/m³; irrigation 0.060-0.160 TND/m³; groundwater 0.112-0.162 TND/m³ — https://www.sonede.com.tn/tout-savoir-sur-leau/la-facture-deau-methodes-de-paiement/...
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